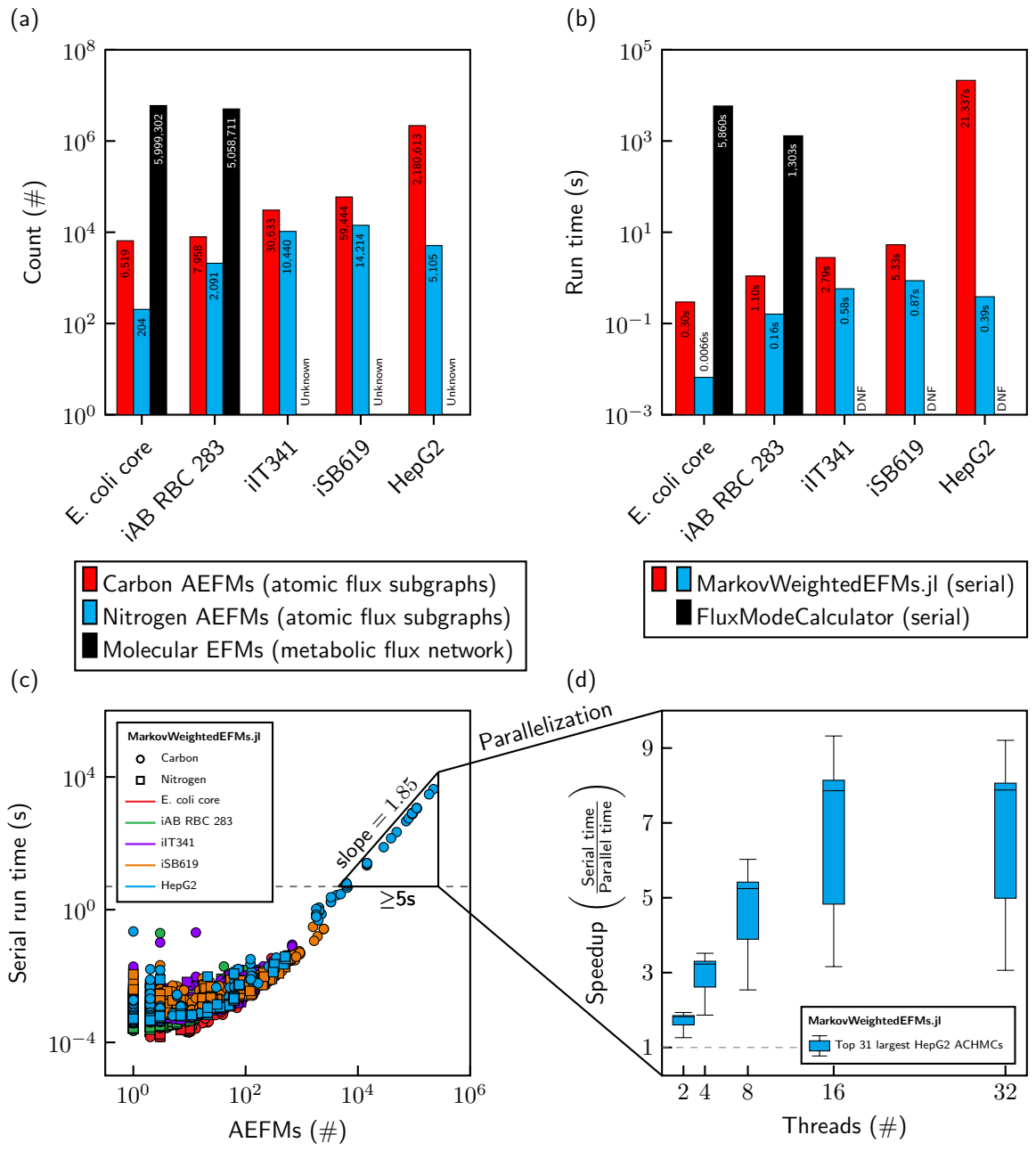
(a)
Count (#)
6,519
204
5,999,302
7,958
2,091
5,058,711
30,633
10,440
Unknown
59,444
14,214
Unknown
2,180,613
5,105
Unknown
E. coli core
iAB RBC 283
iIT341
iSB619
HepG2
Carbon AEFMs (atomic flux subgraphs)
Nitrogen AEFMs (atomic flux subgraphs)
Molecular EFMs (metabolic flux network)
(b)
Run time (s)
0.30s
0.0066s
5,860s
1.10s
0.16s
1,303s
2.79s
0.58s
DNF
5.33s
0.87s
DNF
21.337s
0.39s
DNF
E. coli core
iAB RBC 283
iIT341
iSB619
HepG2
MarkovWeightedEFMs.jl (serial)
FluxModeCalculator (serial)
(c)
MarkovWeightedEFMs.jl
Carbon
Nitrogen
E. coli core
iAB RBC 283
iIT341
iSB619
HepG2
slope = 1.85
≥5s
Serial run time (s)
AEFMs (#)
Parallelization
(d)
Speedup (Serial time / Parallel time)
MarkovWeightedEFMs.jl
Top 31 largest HepG2 ACHMCs
Threads (#)